

NAS10-99001

JOINT BASE OPERATIONS AND SUPPORT

CONTRACT

ATTACHMENT J-1

(CHANGES CURRENT THROUGH MOD 511)

STATEMENT OF WORK

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1.0 PROJECT MANAGEMENT

The contractor shall accomplish the Joint Base Operations and Support Contract (J-BOSC). The J-BOSC provides launch support and public works such as civil engineering services, work control, infrastructure sustainment, and base operations support including information technology, logistics, PMEL laboratories, transportation, airfields operations, and protective, administrative, medical, and environmental services to NASA at Kennedy Space Center (NASA-KSC), the Air Force at Cape Canaveral Air Force Station (CCAFS), at Patrick Air Force Base (PAFB), and at the Florida Annexes.

The contractor shall interact, plan, and effectively coordinate and communicate with all levels of customers, both government and other associate contractors. The contractor shall provide reliable and efficient service that fully satisfies the requirements of the U.S. Air Force (USAF) and NASA and allows them to accomplish their diverse missions without exception and concern for their facilities and services.

The contractor shall implement government initiatives such as the maintenance and restoration of facilities, energy management and conservation, and evolving requirements related to the protection of the environment and minimizing personnel exposure to hazardous materials. Because of constrained budgets, the contractor shall implement trade-offs with other contract functions for the duration of the contract to ensure compliance with regulatory and statutory requirements. Government personnel shall be notified, in a timely manner, of such trade-offs and the impact of these trade-offs to contract requirements.

The contractor shall partner with KSC and the 45th Space Wing to maintain public safety and trust as well as the safety of the workforce. The contractor shall maintain safe and secure operating locations and be flexible and innovative in protecting and preserving physical and environmental assets. The contractor shall prepare for, and effectively respond to, emergency situations and contingencies. The contractor shall support real-time requirements 24 hours a day, 7 days a week.

The contractor shall assure superior customer satisfaction at all stages of work from requirements development through delivery. The contractor shall implement and sustain a customer-oriented work control and planning process that shall provide responsive base support and simultaneously support a multitude of customers having conflicting requirements and priorities.

Management Approach. The contractor shall develop, implement, and maintain a management approach to support NASA-KSC and the Air Force 45th Space Wing's goal to be the premier gateway to space. The contractor shall employ a highly innovative, entrepreneurial, and efficient management program that challenges the status-quo and worker culture in formulating and implementing high quality, timely, and cost-effective base support services. The contractor shall apply an integrated team approach that incorporates a centralized Program Management Office and a single set of consolidated and easily understandable policies and procedures for the J-BOSC program. The contractor shall: take a proactive approach to labor relations involving collective bargaining agreements that includes the formation of a consolidated Labor Management Council; "right-size" the workforce staffing profile; implement an incentive plan for the entire workforce that rewards employees for suggesting and developing ideas that result in quantifiable benefits; and manage the project through incorporation of quality concepts. The contractor shall comply with DRD 1.1-13, Report, Advance Notification of Workforce Reductions, when a workforce reduction is imminent.

The contractor shall optimize J-BOSC support services at the least possible cost within all performance parameters. The organization structure shall balance flexibility and accountability

to ensure responsiveness, product quality, and resource control. The workforce shall be empowered with the responsibility and authority to achieve performance goals.

Business Approach. The contractor shall establish a program which embodies sound financial management concepts that result in affordable costs while continuously improving customer support. The contractor must realize cost savings through improved project efficiency while reducing dependency on government-furnished property and services. The contractor shall team with KSC, the 45th Space Wing, and their customers to optimize resource management. The contractor shall develop a subcontract program that recruits and selects qualified subcontractors, maximizes the use of commercial services, and provides meaningful subcontracting or teaming opportunities to achieve the J-BOSC socio-economic procurement goals.

Resource Management. The contractor shall provide a contract resource management system, to include subcontracts, for the total contract work activity. The contractor resource management system shall provide timely and accurate visibility of contract manpower, cost, and schedule performance and the interrelationship among them. This includes data and supporting variance analyses as well as special exercises involving schedules, work flows, and budgets. Work shall be prioritized, managed, and controlled within program funding levels. Contractor data shall be current, accurate, and complete, and actual costs shall not exceed approved operating plans.

Applicable Work Load Indicators (WLIs) for each WBS are incorporated in attachment J-12 and reported in accordance with DRD 1.1-12, Report and Review, Workload Indicators, Work Backlogs and Deferred Work.

1.1 MANAGEMENT AND CONTROL

1.1.1 Mission Support

The contractor shall support all mission related events per this statement of work.

1.1.1.1 Launch Support. The contractor shall provide all services consistent with mission requirements for support of Space Transportation System (STS) and other vehicle launches. These services shall include planning for operations, maintenance, and logistics support in preparation for launches, operations and maintenance support during launch operations, and requirements following launches prescribed in the Shuttle Integrated Operations and Maintenance Instructions (IOMIs) and Air Force Program Directives. NASA's KPD 8630.3, KSC Shuttle Processing Flight Readiness Certification Review Plan, and ELV/EELV, 501-97 (Draft) Universal Documentation System, depict the standard mission support requirements in accordance with Base Support Policy.

1.1.1.1.1 Launch Readiness Briefings. The contractor shall provide Launch Readiness Briefings for each major milestone of the Space Transportation System (STS) flow. The briefings for the Pad Flow Readiness Review and Launch Countdown Review will be comprehensive in nature, while the remaining briefings will be milestone specific and consist of updates to the comprehensive packages. Joint NASA/SGS Reviews that will be required for STS flow are:

#	Event/Milestone
1	Pad Rollover & Flow Readiness (MAJOR)
2	A5214 – Rollout
3	S00024 – Propellant Load
4	S0017 – TCDT
5	S5009 – Ord. Installation
6	S0037 – Tanking Test
7	Launch Countdown/RTLS (MAJOR)

8	Landing - EOM
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The contractor shall also facilitate at least one JBOSC Launch Readiness briefing for ELV and 45 SW launches inviting the government to attend. The contractor will provide the government with preliminary copies of the launch briefing package to review readiness status prior to the briefing (MOD 436)..

1.1.1.2 24-Hour Turn-Around. The contractor shall provide the necessary resources to support 24-hour turn-around time between launch configurations.

1.1.1.3 Liaison Duties. The contractor shall respond to customer needs 24 hours a day, 365 days a year. The contractor shall perform liaison duties between range users, customers, 45th Space Wing and KSC elements, contractors, and host agencies. Liaison duties include, but are not limited to, receiving support requests, coordinating activities such as work clearance request/excavation permits and utility outages, and passing work requirements to other contractors. Technical Exhibit 5.1-501 lists all the duties. The contractor shall provide a Cape Superintendent function to assist the Cape Commander's office with support activity.

1.1.1.3 Community Special Events. The contractor shall provide miscellaneous support for NASA-KSC, Air Force-CCAFS, PAFB special events, and community special events. This includes the External Relations Office and public affairs support.

1.1.2 Deleted

1.1.3 Flexibility

The contractor shall respond to changing service requirements, including work resulting in unexpected surges, and prioritize activities to best accomplish the intent of the contract in terms of mission support, Cape Canaveral Spaceport Management Office (CCSMO) initiatives, and customer service.

1.1.3.1 PAFB In-Scope Service Requirements. The contractor shall respond to service requirements contained within the general scope of this contract (not specifically identified) at PAFB, upon contracting officer direction, on a non-interference reimbursable basis. The contractor shall document work performed at PAFB under the scope of this paragraph and brief the government on such work on a monthly basis during the Work Management Process IPT. (Mod 306).

1.1.4 Work Classification and Control

The contractor shall properly classify and prioritize all work through the Work Control Center for day-to-day work requirements and through Engineering Services for facilities projects.

1.1.4.1 Schedules and Cost Estimates. The contractor shall provide schedules and cost estimates at the work order level that include cost element breakdowns as requested by the customer.

1.1.5 Work Management System

The contractor shall provide controlled access (network accessible), on-line, interactive automated management and work information system(s). The contractor shall provide authorized users with electronic access to this system and other on-line management information systems and databases. Management and work information systems shall be user friendly and provide required customer data as needed. Access includes the ability to read and download data, and construct and execute ad hoc queries and custom reports with current and

historical data. Data shall be compatible with Microsoft software products or made available through a front-end user interface. Data shall be current, accurate, and complete. The contractor shall develop and maintain user's guides, and provide training for user on how to access on-line Management Information Systems (MIS) and databases.

1.1.6 Automated Systems

The contractor shall provide an information infrastructure that links all applications, resident on all present systems, to share data and information for real-time decision-making across functional area boundaries. The contractor architecture shall provide for interoperability among all management, operations, administrative, and financial systems to improve the efficiency of services and communications for all stakeholders. The architecture shall provide a functional-area independent mechanism to gather data throughout J-BOSC without interfering with the current business process and provide the backbone necessary to sustain orderly systems and applications migration through a set of phases designed to remove system dependencies and improve communication.

The contractor shall provide an on-line commercial off-the-shelf (COTS) work management package to manage scheduled maintenance and repair, corrective maintenance and repair, construction, trouble calls, and other operations and maintenance activities with one integrated tool; ensure work orders are properly handled, necessary approvals are obtained, and follow-on work after emergency mitigation is properly planned and scheduled; improve resource planning and efficiencies across departmental boundaries; capture cost and status tracking at all levels of interest, including individual work orders, by accounting code, project, and WBS area, and feed this information into the customized business system. The purchasing function is centralized through its interface with the customized business system.

1.1.6.1 Government MIS Systems. The contractor MIS shall collect and input required data via Job Ordered Cost Accounting System (JOCAS) submittal in accordance with DRD 1.3-01, Report, Weekly Job Ordered Costs to support government financial reporting requirements.

1.1.6.2 Cost and Schedule Status Reports. The contractor shall provide daily access to an electronic Automated Data Extract Report DRD 1.1-06, Report, Automated Data Extract to keep the government and end customers informed of the cost, schedule, and status of work performed or in progress. Cost data is updated in accordance with DRD 1.3-01, Weekly Job Ordered Costs Report.

1.1.6.3 Data Exchange Standards. The contractor shall support efforts to develop and use data exchange standards (or equivalent functions) to share information concerning Air Force and NASA resources between organizations.

1.1.7 Performance Metrics

The contractor shall develop, maintain, analyze and report performance for contract requirements and provide management reviews. The contractor shall provide objective, measurable metrics as partnered with CCSMO. The contractor shall report existing or potential problem areas with recommended solutions. Applicable Performance Standards for each WBS are incorporated in Attachment J-1, Appendix C and are reported in accordance with DRD 1.1-03, Report, Contract Performance (Metrics).

1.1.8 General Project Management

The contractor shall provide a detailed business model that includes specific recommendations for using the resources base; a practical modernization approach for equipment and facilities; a stewardship approach; J-BOSC cost reductions; an approach for phasing out general purpose

government property; functions to be subcontracted; and a proactive labor relations approach with the unions.

The contractor shall initiate proactive measures to support the government in achieving the goal of becoming the world's premier gateway to space while concurrently maximizing operational effectiveness for the government and commercial customers. The contractor shall form a full partnership with CCSMO to create and implement a beneficial approach to reduce J-BOSC costs associated with relying on government investments in infrastructure and equipment.

1.1.8.1 Risk Management. The contractor shall execute a risk mitigation plan for abating all identified management and technical risks, including labor relations with the unions, loss of skill sets and institutional knowledge to support base operations, information infrastructure, disruption to current service levels, and organizational change necessary to implement the government's vision of the premier gateway to space.

The contractor's Program Manager is responsible for identifying and managing risk elements for the program. While some risk elements include items that are external and beyond the contractor's control, planning for all risks includes proper management recognition and attention to reduce or eliminate these items and their impact. The Program Manager identifies all risk elements and prepares mitigation plans. Each quarter, the contractor's Program Manager shall brief CCSMO on management risk elements and their status concerning potential impact to J-BOSC operations and assess risk elements to be removed or included.

1.1.8.2 Equal Employment Opportunity. The contractor shall submit a quarterly electronic Equal Employment Opportunity (EEO) Report on KSC Forms 32-58 C/G 1/91 including Attachments A, B, and C, in accordance with DRD 1.1-01, Report, Equal Employment Opportunity.

1.1.8.3 Phase-out of Government Property. The contractor shall create and adhere to a disciplined methodology and process to significantly reduce time and costs associated with the analysis and purchase of general purpose property. The contractor shall assess, recommend, and purchase property necessary for maintaining successful day-to-day contract operations. The contractor shall systematically phase out J-BOSC government-furnished general purpose property to support the government's objective of reducing dependency on government-furnished property. The contractor shall maintain a master list of all equipment items purchased for review at final contract termination in accordance with DRD 1.1-11, Report and Review, Replacement of Government Furnished Property.

1.1.8.4 Training. The contractor shall provide a user's guide and training for user access to on-line MIS and databases for NASA, Air Force and other specified organizations and personnel. The contractor shall provide fire and security awareness training. Quality Assurance (QA) training and recurrent training shall be provided to the contractor personnel. Training shall meet customer needs and shall be conducted in a timely and courteous manner. The contractor shall maintain current data for all required certifications. The contractor shall ensure that there are no violations resulting from lapses in required personnel certifications.

1.2 SAFETY AND HEALTH

The contractor shall partner with NASA-KSC and the 45th Space Wing to maintain public safety and trust as well as the safety of the workforce. The contractor shall maintain safe operating locations and be innovative in the protection of personnel and property. The contractor shall establish, implement, and maintain a comprehensive safety and health program that meets current: Programmatic requirements, federal law, and nationally recognized consensus standards referenced within OSHA, NASA guidelines, handbooks, specifications, and standards, and is effective in the identification and mitigation of risks. The safety and health

program shall be fully documented and auditable and the contractor shall support government audit and surveillance activities of contractor plans, procedures, and processes when deemed necessary by the government. Contractor internal assessments, results, and surveillance activities shall be made available and auditable by the government. The contractor shall comply with safety and health requirements set forth in EWR 127-1 Eastern and Western Range Safety Policies and Processes; KNPR 8715.3(T), KSC Safety Practices Procedural Requirements and programmatic requirements. **(Mod 334)**

1.2.1 Safety Compliance

The contractor shall be responsible for providing a safe and healthful work environment and shall perform all operations in a safe manner. The contractor shall establish and implement a safety program that provides protection from injuries to contractor personnel, government personnel, members of the general public, and damage to government facilities, systems and equipment as a result of the contractor's action or inaction. The safety program shall comply with all applicable Federal, NASA, Air Force, KSC and CCAFS regulations and requirements.

1.2.1.1 Operational Safety. The contractor shall develop and implement a process for assessing hazards of contractor activities and protecting personnel and property from exposure to these hazards. The process shall provide for appropriate hazardous operation surveillance, review of hazardous procedures, and assessment of risks associated with deviations from procedures or safety and health requirements.

The contractor shall conduct and document an operational safety assessment of all high risk, first time operations and submit the assessment to the government for review.

For contractor-controlled operations involving hazardous materials that meet or exceed the threshold specified in 29 CFR 1910.119, Process Safety Management standard, the contractor shall develop an auditable program that complies with all of those requirements. All personnel engaged in potentially hazardous operations or hazardous material handling shall be certified as capable to operate the equipment and perform their jobs in a safe manner. Hazardous operation safety certification is required for those tasks that potentially have an immediate danger to the individual (death/injury to self) if not done correctly, or could create a danger to other individuals in the immediate area (death or injury), or are a danger to the environment. All employees shall be trained and/or maintain certification per the requirements outlined in specific government standards/requirements.

1.2.1.2 Institutional Safety. The contractor shall conduct safety inspections of all contractor-occupied facilities on a quarterly basis. The inspections shall be formally documented and all findings tracked to closure. Qualified safety professionals shall perform inspections at least annually.

All injuries, damage and close calls resulting from contractor activities shall be investigated and have appropriate corrective action taken. The contractor shall electronically enter all mishaps meeting the criteria for Type A through Type D mishaps as defined in KNPR 8715.3(T), KSC Safety Practices Procedural Requirements into the NASA mishap-reporting database. All close calls with the potential to have been Type A through Type D mishaps shall also be entered. Mishap reporting methods and timelines shall be in accordance with DRD 1.1-04, Report Anomaly. **(Mod 334)**

The contractor shall establish and implement a construction safety program to ensure that contractor and subcontractor construction operations comply with appropriate requirements and are performed in a safe manner. This program shall include weekly site safety surveillance of all active construction sites awarded by the contractor by qualified safety personnel, documentation and mitigation of all identified hazards, a process to ensure that all construction personnel have

all required training and certifications, a process to coordinate with other KSC and CCAFS organizations that could be negatively impacted by the construction activity, and review of subcontractor safety plans.

The contractor shall provide safety support for the Spaceport public events and visitor programs under the contractor's control. This includes ensuring that facilities, systems, equipment, and services provided by the contractor are safe and ready for use prior to the public event, that traffic control and parking plans are adequate and do not pose any hazards to vehicles and pedestrians, and that any equipment leased, rented, or provided by the contractor from off-site locations is properly assembled and safe for use.

In the event that the contractor determines the need to vary from a government requirement, a formal request must be processed through the appropriate variance review and approval system. The contractor shall perform risk assessments for all variances requested to safety procedures and processes. The risk assessment shall be performed to a managerial level to assure the safety of personnel and equipment is not compromised, and alternate work procedures as a result of a variance shall be evaluated prior to implementation. Requests for a variance to safety requirements (where the only justification is to meet schedules or cost) will be denied.

The contractor shall continue participation in the OSHA Voluntary Protection Program (VPP) as a STAR Site.

1.2.1.3 Transportation Safety. All motor vehicles, railcars, boats, and others used for transportation of personnel or hazardous material shall comply with laws regarding inspections and markings and must have passed an inspection to ensure that the vehicle or vessel is in safe mechanical operating condition. The mode of transportation shall be inspected to the applicable standards of the Federal Highway Administration, U.S. Coast Guard, Department of Transportation (DOT), and Federal Railroad Administration. All vehicles transporting hazardous materials on government and public roadways shall display all DOT required placards, lettering, or numbering. All personnel operating such vehicles shall receive the appropriate training and or licensing. The contractor shall provide support for the Spaceport public events and visitor programs under their control. This includes ensuring that traffic control and parking plans are adequate and do not pose any hazards to vehicles and pedestrians, and ensuring that any equipment leased, rented, or provided by the contractor from off-site locations is properly assembled and safe for use.

1.3 FINANCIAL MANAGEMENT

The contractor shall employ best business practices and sound financial management in the performance of financial management and reporting. Flexible and innovative procedures designed to ensure compliance with the variety of cost charging and reporting requirements of the NASA and Air Force based on category of customer and sources of funds are critical. Complete, accurate, and timely reporting of planned and actual costs is of paramount importance.

1.3.1 Financial Management System

The contractor's financial management system shall be fully integrated with an automated work management system capable of maintaining complete cost integrity, while capturing and reporting cost at the work order level. The contractor shall provide an automated, network accessible, on-line, ad hoc query capability to permit government users access to determine the cost, schedule, and status of work at the level of detail reported in the Job Ordered Cost Accounting System (JOCAS) and 533 reports. The system shall be capable of linking both the accounting and work management systems. The contractor shall notify the government

immediately upon detection of significant errors in their accounting system affecting 30% of work orders and/or 10% of costs reported in one period.

1.3.2 Cost Reports

The following cost reports are required:

- **Job Ordered Cost Accounting System Reports.** The contractor shall submit an electronic Weekly Job Ordered Costs to the Air Force standard JOCAS while maintaining the capability of identifying costs to a daily level. Strict observance of Air Force program funding year of costs is essential in accordance with DRD 1.3-01, Report, Weekly Job Ordered Costs;
- **Geographic Economic Impact Report.** The contractor shall submit an electronic Geographic Economic Impact Report in contractor format by 15 October of each contract year in accordance with DRD 1.3-02, Report, Geographic Economic Impact;
- **Monthly 533 Reports.** The contractor shall submit detailed and accurate Monthly Financial Management Analysis Reports on Forms 533M in accordance with instructions in NPR 9501.2D, NASA Contractor Financial Management Reporting in accordance with DRD 1.3-03, Rev. C, Reports, Contractor Financial Management Analysis, 533M;
- **Cost Pool Analysis.** The contractor shall submit quarterly Cost Pool Report and Analysis of all cost pools to include planned and actual cost pool content on a government fiscal year 12-month time phased schedule, the planned and actual percentages and basis of distribution for the same period, and recommendations for modifications of the rates of distribution. Included as an attachment to this report shall be a detailed dictionary containing a complete description of each cost pool, a detailed listing of the content, and the methodology for the distribution algorithm in accordance with DRD 1.3-04, Rev. A, Report, Cost Pool/JOSA Analysis;
- **Estimate at Complete (EAC).** The contractor shall submit a detailed Estimate at Complete (EAC) at the agency/customer/fund source level for the government fiscal year in accordance with DRD 1.3-05, Rev. B, Report, Estimate at Complete (EAC);
- **Contractor Commitment Cost Data Report.** The contractor shall submit a Contractor Commitment Cost Data Report on the same basis and schedule as the JOCAS cost report in accordance with DRD 1.3-07, Rev. A, Report, Contractor Commitment Cost Data;
- **Launch Scrub Costs.** The contractor shall submit a Launch Scrub Costs Report in accordance with DRD 1.3-08, Rev. A, Report Launch Scrub Costs;
- **Catalog of Services.** The contractor shall submit a Catalog of Services in accordance with DRD 1.3-09, Rev. A, Catalog of Contract Services with Estimated Costs;
- **Job Order Estimates.** The contractor shall submit complete, detailed, accurate, and timely estimates of cost at the agency, customer, and/or Job Order Number (JON) level for current and future government fiscal year requirements in accordance with DRD 1.3-10, Job Order Estimates;
- **Reconciliation of Incurred Costs to Reported Costs.** The contractor shall submit a monthly reconciliation of the costs reported in its general ledger system to the costs reported to the government in either the 533 or JOCAS reports in accordance with DRD 1.3-11, Report, Reconciliation of Incurred Costs to Reported Costs;
- **Contract Operating Plan.** The contractor shall develop and manage an annual contract operating plan by government fiscal year on the basis of agency, customer, fund source, WBS, and cost element, with monthly execution and variance analysis to ensure total contract costs have been reconciled.

- **Monthly Analysis of Actual to Planned Costs.** The contractor shall provide a monthly analysis of actual to planned costs at the WBS level to include volume and rate analysis. The contractor shall partner with the government on the development of variance explanations to ensure common understanding and reporting in accordance with DRD 1.3-12, Monthly Analysis of Actual to Planned Cost;
- **Prime and Sub-contractor Contract Value (CV) Status Report and Review.** The contractor shall provide a Prime and Sub-contractor CV Status Report and Review in accordance with DRD 1.3-13, Rev. A, Report and Review, Prime & Subcontractor Contract Value Status;
- **Direct and Indirect Rates Report and Review.** The contractor shall provide a Direct and Indirect Rates Report and Review in accordance with DRD 1.3-14, Report and Review, Direct & Indirect Rates;
- **Reconfiguration of Contract Value into Cost Structure Reports Response and Review.** The contractor shall provide Reconfiguration of Contract Value into Cost Structure Reports Response and Review.

1.3.3 Job Ordered Cost Accounting System

The contractor shall employ a full job ordered cost accounting system to capture the full costs to the final cost objectives. The final cost objective is defined as the JON. Job Order Numbers may be assigned to specific agency, programs, projects, events, and/or fund sources. Accuracy of cost reporting to the correct JON and government-funding year is critical. Procedures shall be developed to insure adequate processes, training, and/or instruction to contractor work force to ensure maximum accuracy of cost reporting. The contractor shall provide an on-line system capable of identifying and tracking erroneous cost reporting through satisfactory completion.

The contractor shall partner with the government to develop algorithms, as necessary, to distribute indirect costs to appropriate service pools and/or direct job order number. Indirect costs may include costs of common support to each agency, such as Joint Operating Support Agreements (JOSAs), functionally specific common costs (i.e. 3rd level Work Breakdown Structure [WBS] costs which cannot be specifically attributed to the final cost objective), other WBS intersects which bear costs indirectly related to the support being provided.

The contractor's cost accounting system shall have adequate internal checks and balances and audit steps built in to isolate and identify erroneous or incomplete data and procedural deviations under controlled conditions and ensure timely and proper corrective actions consistent with the complexity of the situation. The contractor's system shall employ a point of entry edit capability sufficient to minimize the opportunity for erroneous data to be recorded.

All data reflected in the database shall be adequately supported by auditable documentation that is cross-referenced to provide readily traceable audit trail(s). The system shall contain controls and edits to ensure that all directly identifiable costs are properly identified and reported.

1.3.4 Cost Distribution

The contractor shall update the cost accounting system database so that the status of total costs to a cost objective or final cost objective can be determined. The total costs shall include the direct costs, allocable portions of overhead, G&A, Fee and the distributed indirect costs. For the purposes of overhead, the government will identify some portion of contract direct costs as directly allocable to the overhead.

1.3.5 Budget Development

The contractor shall respond to requests for an annual Operating Plan, NASA's Program Operating Plan (POP), 45th SW Financial Plan (FinPln), and the Air Force Program Objective Memorandum (POM) budget development and other special budget exercises as requested. The contractor shall assist the Air Force and NASA staff in adjusting requirements to meet expected funding availability by identifying the highest priority needs and specifying the operational impact of leaving lower priority requirements unfunded.

1.3.6 Reimbursement Policy

The contractor shall comply with the requirements of the 45th Space Wing Instruction 65-601, Reimbursement Policy when advising the CCSMO of direct costs for Range users that can be identified readily with the particular program support.

1.3.7 Negotiated Estimated Cost (NEC)

The contractor shall ensure complete reconciliation of the Operating Plan to the Annual NEC, and the actual cost to the NEC on the same six month schedule as the accounting calendar.

1.4 QUALITY AND MISSION ASSURANCE

The contractor shall establish, implement, and maintain a comprehensive quality and mission assurance (reliability, maintainability, and quality assurance) program that meets programmatic requirements and is effective in the identification and mitigation of risks. The contractor shall establish and maintain a configuration control process for all configured Facility, Systems and Equipment (F/S/E) that is within the contractor's control for design and operations and maintenance as defined in the Configuration Management Data System (CMD5). Risks shall be identified in sufficient time to allow correction or acceptable risk mitigation without programmatic impact.

1.4.1 Quality Approach/Continuous Improvement

The contractor shall establish and implement a quality management system for all services provided under this contract that complies with the guidance in ANSI/ASQC/ISO 9001 2000 and KNPR 8720.1 **and KNPR 8730.2**. The contractor shall assess and benchmark internal processes to improve services and processes to optimize the delivery of services or products to the customer. The approach shall include a disciplined methodology to determine process effectiveness (time or cost savings), a quality improvement plan and implementation schedule for identified deficiencies, and documentation of lessons learned. (MOD 436) **(Mod 493)**

1.4.1.1 Product and Service Quality. The contractor shall identify the requirement for Government Source Inspection for procurements based on the criticality (1, 1R, and 1S) (reference NSTS 22206 for criticality definitions) of the procurement for ground support equipment, cryogenic/hypergolic propellant mission essential equipment, and other procurements identified by the government based on system interface and vendor performance history.

The contractor shall provide and maintain Acceptance Data Packages, test results, analysis reports, inspection records, and delivery logs in accordance with programmatic requirements for hardware, software, and commodity delivery, or transfer to the government. The contractor shall collect and compile data and information to demonstrate that the products and services delivered to the government are in compliance with programmatic requirements and specifications contained within those programmatic documents.

1.4.2 Systems Safety

All F/S/E of the following types that will be turned over to the contractor for O&M, that have either been temporarily turned over to the government for modification or repair or are a part of

new construction shall be verified/validated through a partnered process as operational to support required mission milestones: (Mod 481)

- F/S/E that is deemed a “critical system” per NSTS 22206
- Non-Critical/MEE F/S/E which have a Combined ORMSD/SAA Assessment (COSA) per Technical Exhibit 2.1.2-003
- Non-critical high/medium voltage F/S/E (for the purpose of ensuring that modifications to the non-critical portion of the system do not impact the critical portions of the system).

System verification/validation shall be accomplished through the following when the situation exists as specified below (excluded from this effort is any normal maintenance that does not alter the configuration):

- The partnered Institutional Design Certification Review (IDCR) process – This process will be used at project completion or at the end of major project phases when all required documentation is available.
- The partnered Interim Readiness Assessment (IRA) process – This process will be used in the interim stages of a project (i.e. construction, activation, turnover, etc.) to document mission support readiness

System verifications/validations (IRA/IDCR) shall include all modifications categorized as “first use” for affected milestones. The process is intended to document that the Appropriate Organizations (i.e. Engineering, Reliability, NASA Center Operations, NASA Safety, etc.) have reviewed the project and determined:

1. In the case of an IRA that either:

- The project content performed is a like-for-like replacement and the configuration documentation matches the modified system. As such, there is no additional risk to the system or the program OR
- The project content performed does impact configuration documentation but the updates are administrative in nature and can be performed after the planned mission OR
- The project content performed does impact configuration documentation and a risk assessment shall be performed to determine if the configuration adds any additional/presently unknown risk(s) to the system or program.
 - o The project open items deemed necessary for mission support shall be listed.

2. In the case of an IDCR that either:

- The project content performed is a like-for-like replacement and the configuration documentation matches the modified system. As such, there is no additional risk to the system or the program and the finalized documentation will reflect this OR
- The project content performed does impact configuration documentation, but in an administrative nature. Updates will be made to the finalized documentation to reflect this OR
- The project content performed does impact configuration documentation and a risk assessment shall be performed to determine if the configuration adds any additional/presently unknown risk(s) to the system or program. The finalized documentation will be updated to reflect this.

The contractor shall prepare or update any required operations and maintenance documentation per the partnered processes. (reference SOW paragraph 2.1.2.4.1).

For JBOSC controlled projects, the system verification/validation and associated OMD required per the partnered processes shall be completed in sufficient time to allow correction or acceptable risk mitigation without programmatic impact. The verification/validation and OMD shall be completed through technical review per the partnered processes with the customer prior to its use in a critical operation or flow. For projects not under the control of the JBOSC contractor, system verification/validation and associated OMD required per the partnered processes will not be completed until the project contractor provides detailed drawings and equipment manufacturer specifications to the JBOSC Activation and Turnover group. System verification and OMD completion dates will be negotiated for each project with NASA-KSC safety and center operations.

In general, completion of documentation updates prior to STS launches will be the priority. Changes in the system configurations requiring interim conditions to be documented will not require complete documentation suite (SAA and OMRSD) updates. The contractor will coordinate with NASA-KSC safety and center operations for overall priorities to accomplish the launch mission and work to a published (and government coordinated) schedule. (MOD 436)

The contractor shall develop and implement a process to identify critical F/S/E which supports the institutional program per the requirements specified in NASA STD 8719.7. For new or modified F/S/E that the contractor builds, procures, or assumes operation or maintenance responsibility for, the contractor shall submit a project specific Hazard Analysis Data Sheet for the government's review. This requirement applies to F/S/E that is critical, mission essential, or has significant or unique hazards associated with it. The format of the Hazard Analysis Data Sheet shall be determined by the contractor and shall include, at a minimum, the type of analysis to be performed, an analysis schedule with established milestones, methods of analysis, depth of effort, identification of single failure points, and proposed mitigation of risks. The plan shall also address reviews and interfaces between the customer and the contractor. The plan shall be developed for the complete lifecycle of the F/S/E, and facility system safety activities shall take place concurrent with the normal facility acquisition process.

For the Shuttle Program, the contractor shall develop and implement a process to identify critical F/S/E which supports the Shuttle Program per the requirements specified in NSTS 22206 and submit a Hazard Analysis Data Sheet for the government's review. For F/S/E identified as Critical, a Failure Modes Effects Analysis /Critical Items List (FMEA/CIL) and a Hazard Analysis shall be performed and Hazard Report(s) generated per the requirements specified in NSTS 22254. The contractor shall present all Critical Items (1, 1R, 1S, and 2) identified in the FMEA and Hazard Report for initial approval to the KSC Shuttle Program Risk Review Board (RRB), and final approval will be granted by the Kennedy Safety Review Panel (KSRP) and the Program Requirements Control Board (PRCB) per the requirements specified in NSTS 07700 Volume V, Information Management, and shall be maintained for all critical items per the requirements specified in NSTS 07700-10-MVP, Shuttle Master Verification Plan. The contractor shall obtain NASA Safety Engineering concurrence(s) on all analyses performed for the SSP. (MOD 436)

1.4.2.1 Reliability and Maintainability. Shuttle System reliability shall be performed under the guidance of NSTS 5300.4 (1D-2). The contractor shall develop and implement a process which ensures the reliability and maintainability throughout the lifecycle of the F/S/E for which the contractor is responsible per the CMDS listing. The process shall include reliability and maintainability assessments for baseline allocations, trend analysis of materials and parts in support of operational integrity, and participation in failure reviews. The assessments shall identify critical items and the operational impacts of associated failure modes. The contractor shall prepare and maintain the associated FMEA/CIL and shall integrate the results of the reliability assessments with the system safety function. The contractor's process shall be based

on the fundamental reliability and maintainability concepts and principles described in NASA-STD-8729.1. (MOD 436)

1.4.3 GIDEP

The contractor shall participate in the Government/Industry Data Exchange Program (GIDEP) in accordance with the requirements of the GIDEP S0300-BT-PRO-010 and S0300BU-GYD-010. The contractor shall review all Failure Experience Data GIDEP ALERTS, GIDEP SAFE-ALERTS, GIDEP Agency Action Notices, and NASA Advisories to determine if they affect the contractor products produced for NASA. For GIDEP ALERTS, GIDEP SAFEALERTS, GIDEP Problem Advisories, GIDEP Agency Action Notices, and NASA Advisories that are determined to affect the program, the contractor shall take action to eliminate or mitigate any negative effect to an acceptable level. The contractor shall generate the appropriate failure experience data report(s) (GIDEP ALERT, GIDEP SAFE-ALERT, GIDEP Problem Advisory) in accordance with the requirements of GIDEP S0300-BT-PRO-010 and S0300BU-GYD-010, and KNPR 8715.3(T), KSC Safety Practices Procedural Requirements, whenever failed or nonconforming items, available to other buyers, are discovered during the course of the contact. **(Mod 334)**

2.0 PUBLIC WORKS

2.1 ENGINEERING SERVICES

The contractor shall provide engineering services to accomplish base operations and support services in support of facilities planning, real property management, facility programming, design services, specifications, engineering documentation, construction management, surveying, cost analysis, facility activation, and energy management at KSC, CCAFS, and the Florida Annexes. For specific Air Force services, please refer to the subparagraphs below. (Mod 451) The contractor shall maintain Air Force and NASA facilities in a cost-effective manner that protects and preserves investments. The contractor shall provide engineering services to NASA in accordance with NASA-STD-8719.11 and NPG 8820.2E.

2.1.1 Facilities Planning

The contractor shall provide a full range of facilities planning services for KSC, CCAFS, PAFB, and the Florida Annexes, as described in the following paragraphs. The facility inspection services that support the long term planning services will not be provided for the Air Forces's F/S/Es. The contractor shall be committed to preserving government assets and protecting the environment. The contractor shall continuously promote the stewardship of government resources among managers and the workforce. The contractor approach shall be to protect the government's investment through dedicated maintenance and repair of infrastructure, advocacy for capital improvements, effective land use for all activities and tenants, and continuous observations and recommendations to appropriate government personnel in all areas. **(Mod 397)**

2.1.1.1 Real Property Management. The contractor shall maintain real property records and perform physical inventories to include utilization of government-accountable facilities located on KSC, CCAFS, and Florida Annexes as listed in Technical Exhibit 2.1.1.1-001, NASA Real Property and Technical Exhibit 2.1.1.1-002, AF Real Property. Upon land withdrawal or return from the Fish and Wildlife Service, the contractor shall enter data in the Cape Canaveral Spaceport Geographic Information System (CCSGIS).

The contractor shall support management of NASA accountable real property in accordance with NPD 8800.14B, Policy for Real Property Management for NASA. The contractor shall

support management of Air Force-accountable Real Property information in accordance with Air Force 32 Series publications 9001, 9002, 9003, 9004, and 9005.

The contractor shall maintain reports, to include the KSC Annual 1400 report, the NASA Quarterly Real Property report, the KSC Trailer Abatement Report, and the Air Force Quarterly Report for Stuart P. McKinney Homeless Act.

2.1.1.2 Space Planning. The contractor shall provide space assessments for NASA-KSC and Air Force facilities, in accordance with KHB 1200.1E and 45 SW Instruction 32-1007.

The contractor shall maintain space utilization information, other than Air Force facility floor plans, for facilities as listed in Technical Exhibit 2.1.1.1-001, NASA Real Property and Technical Exhibit 2.1.1.1-002, AF Real Property (Mod 451).

The contractor shall maintain and validate a current list of all facility managers and serve as the coordinating agency for dissemination of information, training, and instructions for NASA-KSC facilities. The contractor shall provide initial training for all newly assigned facility managers for NASA-KSC facilities. (Mod 451)

2.1.1.3 Comprehensive Master Planning. The contractor shall develop, publish, and distribute the Basic Information Guides (BIGs) for KSC, CCAFS, and the Florida Annexes.

The contractor shall maintain the Air Force General Plan, and the KSC Master Plan in consideration of the Cape Canaveral Spaceport Master Plan (CCSMP) and in accordance with AFI 32-7062 and KHB 1200.1E. The contractor shall document amendments and exceptions to the CCSMP per a government approved process. The contractor shall utilize CCSGIS for maps contained within the CCSMP, KSC Master Plan, and Air Force General Plan. Plans shall be provided in printable format, and electronic format via the intranet. The contractor shall develop, and maintain Air Force Airfield Waivers. The Operational Risk Management (ORM) documentation to accompany airfield waivers will be in accordance with UFC3-150-1 and AFPAM 90-902. The ORM documentation will be worked as a type 3 C project. This includes maintaining the Air Force "On-Installation Obstructions to Airfield Criteria" (Map E-1) and the "Approach/Departure Zone Obstruction" (Map E-2). The contractor shall maintain the Cape Canaveral Spaceport (CCS) Quantity Distance (QD) Map and CCS Line of Site (LOS) Map. The contractor shall develop and process site-plans and shall coordinate the review and concurrence by applicable offices within KSC and CCAFS. The contractor shall review excavation permits for confirmation of valid site plans. The contractor shall provide comprehensive planning studies. The contractor shall provide cartographic support to provide comprehensive map packages.

The contractor shall have capability to provide Area Development Plans. These will be worked as Type 3 C projects.

The contractor shall have capability to maintain the bi-annual Technical Summary of Infrastructure Books, (worked as type 3 C projects) to include the following sections:

1. Electrical Power Distribution System
2. Water Distribution System
3. Wastewater Collection and Treatment System
4. Heating, Ventilation and Air Conditioning
5. Fire Protection Systems
6. Roads & Bridges
7. Structures
8. Airfield

9. Propellants (NASA Only)

The contractor shall have capability to facilitate the implementation and comprehensive use of the 45th Space Wing Facilities Excellence Plan (FEP) by reviewing, enforcing, and coordinating FEP issues. These will be worked as type 3 C projects.

2.1.1.4 Facility Project Programming. The contractor shall perform programming of facility construction projects, real property maintenance projects, and programming for facilities/systems at KSC, CCAFS, PAFB, and Florida Annexes.

The contractor shall maintain SRMC and MILCON project data for the Air Force Automated Civil Engineering System for Project Management (ACES/PM). The contractor shall develop initial project electronic records, monitor and update data as projects develop, and ensure accuracy and consistency of project data.

The contractor shall support annual program calls such as the NASA Program Operating Plan (POP), Construction of Facilities (CoF) Call, KSC Reinvestment Plan, Air Force Facility Investment Metric (FIM) Validation, Air Force Demolition Program Call, and Air Force MILCON.

The contractor shall perform Program Submittal and support for the Air Force Installation Readiness Report (IRR).

The contractor shall prepare specific formal project approval and support documentation for submission to the respective government approval authority in accordance with specific program guidelines. The contractor shall prepare DD Form 1391 and supporting documents for Air Force projects and NASA Form 1509/1510 for NASA projects. The contractor shall prepare other necessary documents, spreadsheets to support various calls, as required.

The contractor shall perform facility project support in accordance with AFI's 32-1021, 1022, and 1032.

2.1.1.5 Geographic Information System (GIS) Support. The contractor shall maintain the (CCSGIS) in accordance with PLP-MP-RP01, CCSGIS Maintenance Plan.

In addition, the contractor shall identify and propose GIS applications for enhancements of Spaceport utilization of GIS.

2.1.2 Design Engineering Services

2.1.2.1 Engineering Design. The contractor shall provide design services (in-house and subcontract) in support of KSC, CCAFS, and the Florida Annexes. The contractor shall prepare designs in accordance with the 45th SW Facilities Excellence Plan (FEP)

The contractor shall provide engineering and drafting services in support of the facilities, systems, equipment and utilities within the contractor's responsibilities. Support shall consist of a skill mix that provides elements such as engineering design, drafting, and cost estimates, to assist in implementing projects to maintain and enhance the performance of J-BOSC assigned facilities, systems, equipment and utilities as identified in the Operations, Maintenance, Engineering, User (OMEU) Matrix.

The contractor shall provide the capability for on-call dedicated engineering and drafting support for high priority and short suspense special projects, and shall accelerate the design program for these projects when necessary.

The contractor shall initiate, as applicable and appropriate for the task, value engineering and constructability analysis during the design phase, provide engineering assistance during construction/installation, and support facility activation following construction/installation.

The contractor shall perform professional engineering services necessary for technical consultations, planning, studies, investigations, surveys, project books, drawings, and specifications. These services involve architectural, chemical, civil, control systems, electrical, environmental, fire protection, mechanical, and structural engineering efforts in varying degrees for assigned projects. Professional Engineers and/or Registered Architects shall seal drawings on an as needed basis.

2.1.2.2 Specifications-Kept-Intact. The contractor shall manage the NASA standard master guide specifications, Specifications-Kept-Intact (SPECSINTACT), for the preparation of facility construction project specifications. The contractor shall produce updated text changes to the NASA and KSC Mastertext for electronic transfer to the National Institute of Building Science (NIBS) twice annually; produce and maintain the NASA Mastertext database for use by the SPECSINTACT user community; maintain the local KSC SPECSINTACT guide specifications, provide review and engineering support, and update the Specification Documentation System (SDS) computer to ensure that the latest guide specifications are available; provide engineering support to users; and support the project with involvement in NIBS and the Construction Specifications Institute (CSI). The contractor shall convert government-provided Air Force Mastertext to the SPECSINTACT format and post to the SPECSINTACT web page for use by the SPECSINTACT user community and electronic transfer to the National Institute of Building Science (NIBS) twice annually. The contractor shall provide engineering support to users. The contractor shall support the project with involvement in NIBS and the CSI.

The contractor shall bi-annually provide the NASA SPECSINTACT Configuration and Change Control Board (NS-CCBB) with information on pertinent industry news, engineering updates, and budget requirements for review and action. The contractor shall continuously review and update the KSC SPECSINTACT database to latest industry reference standards, federal regulations, and executive orders, and shall ensure that all sections are reviewed and updated with the latest technical content at least once every 3 years. The contractor shall disseminate current products and information.

The contractor shall perform tasks associated with maintaining the SPECSINTACT computer software in support of NASA, the Army Corps of Engineers, the Air Force, the Naval Facilities Engineering Command, and any other future government agency and their architectural/engineering firms. The contractor shall perform all software tasks associated with supporting member agencies guide specifications, including analysis and incorporation of software change requests, referenced publications, and format. The contractor shall prepare recommendations to be considered by the SPECSINTACT Interagency Configuration Control Board (SICCB). The contractor shall support all NASA and Interagency Control Board meetings, NIBS meetings, and SPECSINTACT presentations, and utilize and maintain a web site for software, text, and documentation distribution. The contractor shall bi-annually arrange a SICCB meeting and provide its members with information on pertinent industry news, software update, and budget requirements for review and action.

2.1.2.3 Engineering Programs.

2.1.2.3.1 J-BOSC Managed Pressure Vessel/Systems (PV/S). The contractor shall manage and implement the PV/S In-service Inspection (ISI) process in accordance with KNPR 8715.3(T), KSC Safety Practices Procedural Requirements, for vessels/systems as listed in Technical Exhibit 2.1.2-001. **(Mod 334)**

The contractor shall perform any required PV/S Certifications for vessels/systems, as listed in Technical Exhibit 2.1.2-001, to applicable American Society of Mechanical Engineers (ASME) pressure vessel or piping standards.

The contractor shall maintain the NASA PV/S database for J-BOSC PV/S units as listed in Technical Exhibit 2.1.2-001.

2.1.2.3.2 Engineering Inspections.

Shuttle Landing Facility. The contractor shall conduct inspection of the Shuttle Landing Facility (SLF) runway, taxiway, and apron pavements once every 2 years. Inspections shall include a Pavement Condition Index (PCI) survey conducted in accordance with Federal Aviation Administration (FAA) AC 150/5380-6 and AC 150/5320-12C requirements.

KSC Bridges. The contractor shall conduct inspections of KSC bridges, which include the Indian River Bridges (2), State Road 3 Overpass Bridges (2), Banana River Bridge, Banana River Relief Bridge, Jay-Jay Railroad Bridge, and the Haulover Canal Bridge once every two years. Inspections shall be conducted using State of Florida Statute 335.074, Safety Inspection of Bridges.

2.1.2.3.3 Power Systems Analysis (PSA). The contractor shall review, update and maintain engineering analyses and studies associated with critical or mission essential F/S/E. These analyses shall be configured for critical and mission essential F/S/E. The contractor shall provide and maintain a comprehensive PSA (short circuit and load flow) of an approximately 4,200 bus power system that extends from the 115kv power substations to the facility transformer secondary bushing. The PSA shall be updated once per year to include approximately 50 element changes resulting from design modifications or maintenance change outs.

The contractor shall provide and maintain Power System Coordination (PSC) documentation for approximately 3,000 devices within the 115kv power substations to the facility transformer secondary bushing, approximately 40 of which are revised each year due to routine maintenance.

The contractor shall file hardcopy and software computer aided engineering (CAE) files of these studies in the Engineering Documentation Center (EDC).

2.1.2.3.4 Department of Transportation Compliance for Propellants and Life Support Mobile and Portable Equipment. The contractor shall revise and maintain mobile and portable equipment marking plans in accordance with Part 49 CFR for approximately 80 different types of units.

The contractor shall prepare renewal packages for DOT exceptions as listed in Technical Exhibit 2.1.2-002.

The contractor shall review Part 49 CFR, Chapter 1, Subchapters A, B, and C, Chapter III, Subchapters A and B, and determine the effect of applicable changes in the Federal Register.

2.1.2.4 Joint Technical Documentation Control Center (JTDC). The contractor shall maintain the baseline of and prepare, maintain, and deliver to the J-BOSC Engineering Documentation Center (EDC) updated engineering documentation for assigned J-BOSC F/S/E to include (as appropriate to the task) engineering documentation for conceptual, preliminary, and detailed design for sustaining engineering tasks. The contractor shall provide configuration controlled release and retrieval coding for J-BOSC engineering originals and associated engineering drawings, specifications, and documents. The contractor shall provide designated engineering records, document, and data management and control to include documentation closure against configured and critical F/S/E and related systems. The contractor shall prepare, maintain, release, and control documentation (such as OMI and PMI, maintain or "record" drawings as shown in CMDS) required by the contractor to perform operations, maintenance, repair, and modification of all assigned structures, facilities, utilities, and equipment on KSC,

CCAFS, and the Florida Annexes, excluding PAFB. The contractor shall perform engineering documentation support as required to produce the Facility Number Report (CMDS/CID).

2.1.2.4.1 Operations and Maintenance Documentation. The contractor shall modify and update the following Operations and Maintenance Documentation (OMD), which includes drawings, instructions, specifications, System Assurance Analyses/System Criticality Analyses (SAAs/SCAs) and other Risk/Hazard Analysis Packages, SPAs, PV/S and Department of Transportation (DOT) compliance reports, OMRSDs, OMI, and PMIs as directed by the applicable J-BOSC Configuration Control Board (CCB). At a minimum, all NASA critical F/S/E shall require drawings, SAAs/SCAs, SPAs, OMRSDs, OMI, and PMIs. For non-critical F/S/E where OMD may add value to effective system maintenance or reliability, a Combined OMRSD SAA Assessment (COSA) will be generated. The contractor and government shall partner such cases (Reference Technical Exhibit 2.1.2-003. (MOD 436)

The delivery of SAA, SCA, OMRSD documents shall be consistent with Section 1.4.2 "System Safety". The delivery of OMI, PMI and SPA documents shall be consistent with section 2.2.

As-built drawings for NASA maintained drawings, provided by the government or other contractors, shall be processed and released to EDC or logged for document maintenance upon final acceptance of turnover and subsequent receipt of accurate as-builts, and turnover documentation per the following schedule, or as directed by the applicable J-BOSC CCB:

- 95% within 60 days for government critical systems--remainder within 90 days;
- 75% within 90 days for government configured and mission essential systems—remainder within 120 days; and,
- No more than 10 Engineering Orders (EOs) per maintained document, excluding pending EO (not yet implemented) and EOs for Relay Settings (MOD 436)
- * - For AF drawings, applies only to electrical and potable water systems (Mod 451)

For AF Configured drawings, existing drawings are accepted as "Best Available Data" and will be as-built as projects affect the drawings, and only to the extent the project affects or verifies the configuration. Reference Technical Exhibit 2.1.3-001 for the anticipated annual project value of NASA-KSC Construction of Facilities (Coff) projects. (MOD 436)

2.1.2.4.2 Critical/Configured Systems Documentation. The contractor shall provide critical/configured systems baseline identification and change control; coordinate and maintain contractor interface agreements (memoranda of understanding); Configuration Management Data System (CMDS) technical data administration and maintenance; coordination and maintenance of CMDS memoranda of understanding between contractor and others as needed; maintenance of the CMDS configuration identification module; and provision of required CMDS release and retrieval coding.

The contractor shall utilize consistent, unambiguous data classification and maintenance codes to ensure that classifications and criticality status for all F/S/E and associated OMD are clearly identified. Classification codes shall include Critical, Non-Critical, Configured, Mission Essential Equipment, and Maintained items. (MOD 436)

2.1.2.4.3 Data Management. The contractor shall provide data management for CMDS derived web-based F/S/E O/M/E/U Matrix in accordance with DRD 2.2-01, Report, Facilities, Systems, and Equipment Operations/Maintenance/Engineering/User Matrix.

2.1.3 Construction Engineering

The contractor shall perform construction management, construction services, field surveillance and inspection, construction cost engineering, land surveying, facility activation and turnover support, environmental consultation, and associated planning, scheduling, and administrative support for those construction projects implemented by the contractor or implemented by others and assigned to the contractor for construction engineering support.(Mod 451) **(Mod 481)**

The contractor shall provide construction engineering services beginning with environmental consulting and coordination during the project planning and design phase. The contractor shall also provide constructability reviews at the 90% design phase. Support shall continue through the construction, beneficial occupancy, activation/turnover, and warranty program phases. (Mod451) **(Mod 481)**

2.1.3.1 Construction Management. The contractor shall provide project and construction management support for all NASA-KSC facility projects assigned to the contractor for design, construction management, and/or implementation. The contractor shall implement a construction management approach that provides high-quality construction performance and facilities that meet functional requirements on schedule and within approved funds. The contractor shall ensure that all bid packages contain complete and detailed documentation that fully describes each facility configuration, performance requirement and the intent for the facility construction work, including environmental permitting and construction. The contractor shall track, analyze and report project status, scope, schedule, and cost. The contractor shall address these items monthly and report on variances from agreed upon cost and schedule. (Mod 451)

2.1.3.2 Construction Services. The contractor shall provide field surveillance and associated inspection services to ensure successful implementation of facility, equipment and Program projects per Technical Exhibit 2.1.3-001. The contractor shall provide inspection services, while focusing on safe and environmentally sensitive execution of facility, equipment and Program projects. The contractor shall perform daily site visits and witness specified on-site acceptance testing and mandatory inspection points. The contractor shall document non-conformance with drawing, specifications, contract provisions, and safety regulations. The contractor shall prepare and track punch list items to completion. The contractor shall coordinate shop support for outages, system testing, and utility locates. **(Mod 481)**

The contractor shall provide environmental support to NASA facility, equipment and Program projects as required. The contractor shall provide environmental consulting expertise and coordination throughout the project delivery process. The contractor shall provide consultant services at the Project concept, NASA Program Operating Plan, Preliminary Engineering Report, Engineering Design and Project Construction phases. The contractor shall provide input to the identification of potential environmental compliance requirements and documentation of requirements necessary to ensure that environmental requirements are adequately scoped. The contractor shall participate at pre-work meetings to ensure that environmental requirements are communicated to all contractors. The contractor shall supplement inputs from other environmental subject matter experts from NASA, consulting engineers, and various J-BOSC organizations to support the planning and design engineering phases of facility, equipment and Program projects. This support will also include project field implementation consultation for environmental requirements such as waste management, stormwater pollution prevention plans, etc. and the monitoring of field compliance with KSC/regulatory requirements and best practices. **(Mod 388) (Mod 481)**

The contractor shall provide tracking and distribution support for construction contractor shop drawings, material submittals, and other information provided for government review and approval. The contractor shall maintain submittal tracking logs to track the review and approval

process. The contractor shall maintain submittal files as official Government records. The contractor shall maintain tracking logs for construction contract change orders and supplemental agreements. The contractor shall track estimated, proposed, evaluated, and negotiated costs.

The contractor shall have construction inspectors on the job site when needed to support significant construction operations, critical inspection points, acceptance tests, and final inspections.

2.1.3.3 Construction Cost Engineering. The contractor shall provide facilities, systems, and equipment cost estimating and cost engineering services to support NASA-KSC construction activities. The contractor shall provide construction cost engineering and scheduling support for assigned projects. The contractor shall review, analyze, and provide recommendations regarding contractor progress schedules, pay requests, cost proposals, and claims. (Mod 451)

The contractor shall prepare complete findings of facts, including impact assessments for claims and proposed change orders with supporting cost estimate, for use by the Government on those construction contracts under surveillance. Cost analysis and estimating construction support shall be as requested for projects under surveillance commencing no later than project notice to proceed.

The contractor shall review and analyze construction activities in accordance with the following schedule or by the negotiated due date 90% of the time:

- Construction schedule – 3 days
- Pay request – 3 days
- Cost proposals – 10 days
- Claims – 30 days

The contractor shall conduct independent assessments and cost estimation of selected facility, equipment and Program projects, including quality control of project cost estimates. The contractor shall maintain the capability to perform life cycle cost analyses of facility, equipment and Program projects. The contractor shall develop and maintain a project cost element database, including monthly cost indices, for facility, equipment and Program projects. The contractor shall develop reports that analyze project bids and compare the bids against official government estimates. (Mod 481)

2.1.3.4 Land Surveying. The contractor shall provide survey support for verification of requested in-place construction to required elevations and grades for projects assigned to the contractor for construction management services and minor data collection support for customer use. Geographic Information System (GIS) standards shall be utilized to support real estate data for KSC, CCAFS, and the Florida Annexes for the GIS maintenance function.

2.1.3.5 Facility Activation and Turnover. The contractor shall implement a facility activation plan that ensures that new or modified facilities, systems and equipment procured by the government are coordinated, reviewed, and accepted by J-BOSC for engineering, operations, and maintenance responsibility.

In support of facility activation, the contractor shall prepare Interim and Final O&M turnover packages for assigned projects. The contractor shall participate in projects requirement reviews, design reviews, project status meetings, systems acceptance testing, and final inspections. The contractor shall prepare real property turnover packages for government signature and distribute to appropriate KSC/CAFS O&M and Real Property Officers. The contractor shall ensure that all OMD documentation has been properly processed and distributed to responsible O&M support organizations.

Projects implemented by others require that turnover packages be provided to J-BOSC for integration of F/S/E assigned to the contractor for O&M responsibilities.

The contractor shall prepare KSC 21-136 forms and submit interim O&M turnover documentation to allow start of breakdown and code required maintenance within 2 weeks of customer's request after substantial completion. (MOD 436)

The contractor shall prepare and submit final turnover documentation within 30 working days after completion and closure of all punch list items and complete all OMD as required in sections 1.4, 2.1 and 2.2. (MOD 436)

2.1.4 Energy and Water Conservation

2.1.4.1 Energy Management Program. The contractor shall provide program management support for the overall KSC Energy Management Program. The contractor shall conduct an Energy Management Program that plans, implements, and measures the status of conservation initiatives for J-BOSC responsible NASA-KSC F/S/E in accordance with the O/M/E/U Matrix. The contractor shall coordinate an effective utilities conservation program that includes hot and chilled water, natural gas, propane, fuel oils, and electric power. (Mod 451)

The contractor shall perform Facility Energy Audits for all NASA J-BOSC facilities which meet the intent of Executive Order (EO) 13123 and NPR 8570.1, Chapter 4. (Note: Due to a lack of government funds, the contractor will not be required to comply with EO 13123 and NPR 8570.1, Chapter 4 energy audit requirements until such time as funds become available).

The contractor shall actively support the KSC energy program and shall conduct an internal program for J-BOSC. The contractor shall provide the following support to the energy management and conservation program: (Mod 451)

- Support the coordination of the contract-wide program through information, collection and analysis;
- Identify and assist in implementing opportunities for applying cost-effective alternative fuel technologies and energy reduction projects;
- Identify and assist in implementing opportunities for cost-effective purchasing and application of utilities; and
- Utilize Type 3I energy saving projects and energy efficient designs for future projects

2.1.4.2 Energy Conservation. The contractor shall reduce energy consumption as defined in Performance Standard 2.1.4-01 (Energy Use Index in J-BOSC/NASA Facilities). The performance standard shall consist of two measures and shall be posted quarterly on the contractor's energy web page. One measure shall be calculated from standard EUI calculations and is not a contract requirement goal. The second measure shall be calculated without the Orsino Area miscellaneous category and is the contract requirement goal.

2.1.4.3 Reports and Plans. The contractor shall provide energy management program status to the government and shall collect, analyze, validate, and prepare reports and plans as listed in Technical Exhibit 2.1.4-01 – Energy Management Office Periodic Report Requirements. Reference Air Force Energy Program Procedural Memorandum (AF EPPM) 96-3 for Defense Utility Energy Reporting System (DUERS) for Air Force Periodical Report.

2.1.4.4 Water Management Program. The contractor shall provide a water management and conservation program in accordance with EO 13123 for KSC and CCAFS. The contractor shall conduct a Water Program that plans, implements, and measures the status of conservation initiatives for J-BOSC NASA-KSC responsible F/S/E. (Mod 451)

2.2 INFRASTRUCTURE

The contractor shall provide operations, maintenance, engineering, and custodial services for the facilities, systems, equipment, grounds, and utilities at KSC, CCAFS, and the Florida Annexes at **JDMTA, Ft. Pierce, Stuart and Wabasso** to accomplish safe, effective, and environmentally compliant base operations and support services. Assigned F/S/E are designated by the contractor's O/M/E/U Matrix in accordance with DRD 2.2-01, Report, Facilities, Systems, and Equipment Operations/Maintenance/Engineering/User Matrix. **(Mod 472)**

The contractor shall continuously maintain an infrastructure maintenance program based on the Reliability Centered Maintenance (RCM) principles and the Integrated Logistics Support (ILS) requirements. The contractor shall continuously maintain the inventory and systems analysis process and maintain optimal baselines for each system and subsystem. The contractor shall not provide systems analysis process for AF. **(Mod 397)**

The contractor shall develop and provide a Five-Year Facility Maintenance and Facility Project Plan that establishes long-term system sustaining projects and planning in accordance with DRD 2.2-04, Five Year Facility Maintenance and Facility Project Plans. **(Mod 397)**

The contractor shall maintain an electronic F/S/E O/M/E/U Matrix in accordance with DRD 2.2-01, Report, Facilities, Systems, and Equipment Operations/Maintenance/Engineering/User Matrix.

2.2.1 Facilities/Systems/Equipment (F/S/E)

The contractor shall provide a comprehensive integrated management information system for receiving, validating, scheduling, controlling, subcontracting, project management, cost management, and tracking all work associated with the contract. The work control function shall identify all tasks, costs, durations and resources required. The contractor shall identify and implement tasks and work necessary to ensure that all facilities, structures, systems, and equipment for which O&M responsibility is assigned shall continue to function at their original capacity and at their design efficiency. Work to be identified includes preventive maintenance, corrective maintenance, and overhaul or replacement to obtain best value.

The contractor shall provide and maintain an on-line, interactive automated work control system that provides current, accurate visibility of all work. The system shall provide sufficient scheduling and reporting capability to communicate to customers work milestones (programming, design, procurement/contracting, pre-bid, bid opening, award, notice to proceed, completion dates, revised completion dates, construction status, utility locates) and cost (Rough Order of Magnitude (ROM), Final Design Estimate (FDE), original contract, current contract, pending change orders, and final contract) to completion, in accordance with DRD 2.2-08, Type 3C Project and Work Status Report.

The contractor shall provide the government with access to the man-hours expended on government submitted projects. Information input into the system shall include the facility number, system description, equipment number, utility description, and a complete description of work to be done. Information shall be sufficient enough to allow government customers to enter the system and perform searches for any F/S/E. The contractor shall provide the government with F/S/E reports in accordance with DRDs 2.2-04, Five Year Facility Maintenance and Facility Project Plans, 2.2-05, Backlog of Maintenance and Repair (BMAR), 2.2-06, Facility Maintenance Execution Summary, and 2.2-07, Facility and System Availability.

The contractor shall ensure the reliability of assigned facilities, systems, and equipment. Maintenance shall be performed to prevent deterioration to the point at which major refurbishing is required prior to the end of the designed life cycle to restore the system, facility, or equipment.

Reliability Centered Maintenance (RCM) principles shall be employed in accordance with NPR 8831.2. For critical, mission essential, or life safety F/S/E, the contractor shall perform 100% of scheduled preventive/predictive maintenance (PM) tasks including all applicable maintenance/validation steps contained therein. Any PM not performed or completed in its entirety for critical, mission essential, or life safety F/S/E shall be briefed or reported to the responsible government functional subject matter expert (e.g. TA, CES) and to the contracting officer's technical representative (COTR) on a weekly basis. Maintenance analysis decisions requiring changes to the levels of preventive/predictive maintenance, shall be documented on the Maintenance Action Request (MAR) form, FAM-F-0006. Reductions in PM requirements or frequency for any F/S/E (i.e. critical, mission essential, life safety, or other) shall require pre-approval of the responsible government functional subject matter expert. (MOD 436)

Warranty Maintenance. The contractor shall execute warranty provisions and maintain warranty or guarantee records for all equipment and material in the facilities support area during the time they are under warranty or guarantee; investigate the failure of any covered equipment or material and report findings to the appropriate customer; and take no action that would void a warranty without prior approval from the government representative. The contractor shall be responsible for obtaining warranty documents on all warranted equipment and materials installed by the contractor. Warranty documents shall be provided to the contractor upon installation of new equipment by others.

2.2.1.1 Operations, Maintenance, and Engineering. The contractor shall provide operations, maintenance, and engineering for all F/S/E, in accordance with the O/M/E/U Matrix, DRD 2.2-01, Report, Facilities, Systems, and Equipment Operations/Maintenance/Engineering/User Matrix (Technical Exhibit 2.2—017) including associated F/S/E firmware and software. When the contractor, the government, or both have identified F/S/E omissions or deletions, the O/M/E/U Matrix shall be updated by direction from the contracting officer. However, during the update or turnover process, the contractor shall continue to provide operations, maintenance, and engineering services for the pending facilities, systems, subsystems, structures or other items required for function. **(Mod 286)**

For the implementation of maintenance, the contractor shall, at a minimum, have appropriate OMI, PMI and SPAs for all NASA Mission Essential and or Critical Systems. Upon final acceptance of turnover and subsequent receipt of accurate as-builts and vendor documentation, the above maintenance OMDs shall be updated per the following schedule, or as directed by the applicable J-BOSC CCB:

- 95% within 60 days for government critical systems--remainder within 90 days;
- 75% within 90 days for government configured and mission essential systems-- remainder within 120 days; and
- No more than 10 Engineering Orders (EOs) per maintained document excluding pending EO (not yet implemented) and EOs for Relay Settings.

Reference Technical Exhibit 2.1.3-001 for the anticipated annual project value of KSC Construction of Facilities (CofF) projects.

The contractor shall provide problem reporting for all anomalies and off-nominal configuration changes for critical and/or mission essential F/S/E. Problem documentation shall include anomaly description and analysis, critical and/or mission essential F/S/E impacts, operational constraints, troubleshooting steps and findings, and corrective actions performed leading to closure of the anomaly. Problem documentation shall be readily accessible to responsible government functional subject matter experts. All open problem reports shall be briefed to

government management on a regular basis. The formats and frequencies of the reports shall be partnered with the Government. (MOD 436)

2.2.1.2 Deleted.

2.2.1.3 Work Orders. (Mod 244) The contractor shall initiate and/or respond to construction, modification, maintenance and repair work orders as defined in Attachment J-9 and in accordance with the cost Ceiling established in the table below.

	GFY	Internal (Type 3I)	Customer (Type 3C)
Air Force	2004 - 2008	AFI32-1032	AFI32-1032
NASA	2004 - 2005	\$200,000	\$5,000,000
NASA	2006 - 2008	\$200,000	*\$500,000

Table 2.2.1.3-1 Work Order Cost Ceiling

All Air Force type 3 projects (3I and 3C) require Air Force coordination/approval prior to implementation. All projects greater than \$1,500,000 require pre-approval by both the COTR and the Government Contracting Officer. Annual Type 3C Work Order Most Probable Cost Adjustments are defined in Technical Exhibit 2.2.1.3-001. All work requirements shall be documented in accordance with AFI 32-1032 or NPG 8820.2 as appropriate.

*This ceiling may be increased by mutual agreement of the parties to an amount not to exceed \$5M in any fiscal year. (Mod 244)

The contractor shall perform project management, construction management, and subcontracting management, and shall provide Intranet based reporting for government-selected projects and other selected work to include project scope, status, budget, schedule, and accountability in accordance with DRD 2.2-08, Type 3C Project and Work Status Report.

Infrastructure-related work shall be given a numerical priority based on safety, environmental and mission criticality, and/or customer requirements. The contractor shall receive, document, schedule, and control infrastructure work requirements. The contractor's Work Control System shall assign a discrete Work Order Number (WON) for all maintenance, repair, construction and support services.

Every 6 months the contractor Work Control shall notify customers of current open Type 3C work orders and ask customers to revalidate the need for the work. If the customer indicates the work is no longer required, the contractor Work Control shall close those work orders.

Every 3 months the contractor Work Control shall notify the contractor Crafts/Shop management of current open continuous, repetitive or collection (Type 4) work orders to determine if the work is completed. If the work is certified as completed by the contractor Crafts/Shop Management, then the contractor Work Control shall close those work orders.

The contractor shall submit a Type 3C and a Type 3I Work Order Status Report in accordance with DRD 2.2-09, Type 3C Work Order Status Report and DRD 2.2-10, Type 3I Work Order Status Report.

2.2.1.4 Maintenance Support. The contractor shall provide engineering services and studies to ensure the economical maintenance of the CCAFS and KSC infrastructure to support space lift and launch base support missions. The contractor shall provide life-cycle sustaining engineering

for F/S/E. Contractor life-cycle sustaining engineering shall provide continuing engineering support to: maintain a design that fulfills its original design intent and is compatible with the operational use; upgrade operational performance capabilities through product improvement redesign for more cost-effective operations; incorporate approved changes in requirements as they evolve; and provide other engineering support herein.

The contractor shall perform National Fire Protection Association (NFPA) code required maintenance on all fire alarms, fire suppression, emergency generator, emergency power, emergency lighting, and other applicable assigned F/S/E. The contractor shall perform routine and recurring maintenance on all assigned F/S/E as prescribed by maintenance analysis to ensure safe and efficient operations. The contractor shall perform maintenance on a non-interference basis with facility operations. Where this is not possible, outage requests for planned maintenance shall be coordinated with facility operations a minimum of 14 days in advance.

2.2.1.5 Deleted (MOD 436)

2.2.1.6 Deleted.

2.2.1.7 Road Shoulder Maintenance and Clear Drainage Systems. The contractor shall maintain road shoulders free from ruts, washouts, and dead or missing grass. The contractor shall clear drainage systems, including culverts, and control vegetation along access roads to allow waters to flow freely to natural basins or collecting points. Roads, roads shoulders, and drainage systems shown in Technical Exhibit 2.2.1.7-01 and per the schedule defined in the 5-Year Drainage System Maintenance Plan.

2.2.1.8 Preparation for Launch, Test and Training and Search (Xenon) Lights/Generator Support. The contractor shall provide facilities, equipment, and utilities support during pre-launch and landing for all launches according to shuttle integrated operations and maintenance instructions and Air Force Program Directives. NASA's KPD 8630.3, KSC Shuttle Processing Flight Readiness/Certification Review Plan, and ELV/EELV, 501-97 Universal Documentation System, depict the standard mission support requirements. **(Mod 257)**

The contractor shall prepare KSC Pad A&B slidewire bunker areas for launch, test, or training exercises as listed in Technical Exhibit 2.2.1.8, Pad A&B Slide Wire Area Support Standards. Preparation includes: grading area between sidewalks, removal of accumulations of eroded sand from perimeter of areas within the bunkers, and removal of large shells, rocks, and debris, adding clean builders sand as necessary, and removal of sand deposited on bunker sidewalks. The contractor shall inspect bunker dirt cover for proper depth and erosion control. If necessary, the contractor shall replenish dirt and seed properly. Sand fill depth at slide wire landing shall be inspected and sand replenished, if necessary, in accordance to the established safety guidelines.

The contractor shall maintain and operate search lights and associated generators in support of all government launches. This support includes four remote Trans Atlantic Launch Sites (TAL-sites). These sites include Moron, Spain, Zaragosa, Spain, Ben Guerir, Morocco and Istres, France. The Contractor is required to provide manning of three of these TAL-sites per shuttle launch. Existing inventory of search lights (xenon) is 126 units. The existing inventory of generators associated with the search lights is 76 units. In addition to this inventory, the contractor shall be responsible for providing parts for 44 search lights located in White Sands and Edwards AFB. **(Mod 257)**

2.2.1.9 Deleted.

2.2.1.10 Office Moves and Alterations. The contractor shall perform moves and office alterations, including the relocation of personnel and office materials, partitions, and systems

furniture, and perform the necessary design layout. The Launch Operations and Support Contract (LO&SC), Space Program Operations Contract (SPOC), KSC Visitor Complex Concessionaire, and Checkout and Payload Processing Services Contract (CAPPS) are responsible for moves and non-floor-to-ceiling alterations within their assigned areas. **(Mod 402)**
Air Force support may be provided through Type 3C projects. The contractor will only accept tasking for move/mods for Air Force customers on an AF Form 332 and execute only with approval of 45 CES. (Mod 441)

Services extend to minor modifications as necessary. In all facilities, the contractor shall design and implement all necessary convenience outlets, lighting, fire detection and alarm, and HVAC systems to accommodate the move. The contractor shall conduct Safety Design Reviews of Move/Mod Packages.

2.2.1.11 Move/Relocate Portable Facilities. The contractor shall move or relocate all government real property accountable portable buildings, trailers, modulars, and mobile homes identified in the O/M/E/U Matrix. This includes all associated preparation, siting, utilities, and fixtures. The contractor shall perform minor site work including the moving of Modulars, single- and double-wide trailers, and installing tie-downs, steps, blocking, and skirting. After removal of a facility from an area, the contractor shall clean up all remaining debris and grade the site to its original state, including re-sod or resurfacing as necessary. The contractor shall support the implementation of the NASA Trailer Abatement Plan.

2.2.1.12 Corrosion Control and Facilities Painting. The contractor shall provide corrosion control to include support to cathodic protection systems under J-BOSC, per the O/M/E/U Matrix and facilities painting support to include road markings, fabrication, and installation of signs and banners.

The contractor shall provide abrasive blasting and protective coating services for KSC and CCAFS, and to equipment transported to the corrosion control facilities at KSC and CCAFS by CAPPS, Space Station, SPOC and the LO&SC contractors. The contractor shall provide corrosion control and paint services to include scaffolding and rigging support as required, for maintenance activities of assigned structures, facilities, systems, and equipment. The contractor shall prepare and maintain a Cathodic Protection Booklet in accordance with AFM 85-5, AFI 32-1054, and DRD 2.2-11, Cathodic Protection Report. One (1) hard copy and one (1) electronic copy shall be delivered on an annual basis. Painting color scheme of facilities shall be in accordance with facility excellence plans. The contractor shall provide National Association of Corrosion Engineers (NACE) Level 3 inspection services as required. **(Mod 397) (Mod 402)**

2.2.1.13 Deleted.

2.2.1.14 Asbestos Protection Program. The contractor shall manage the asbestos containing material (ACM) program at all contractor maintained F/S/E, as specified in the O/M/E/U Matrix, in accordance with the 45th Space Wing and NASA-KSC Asbestos Management Program. The contractor shall develop and maintain a written asbestos management and operation plan as detailed in Appendix "A" of Section "J", Attachment "J-1" (to include a 5-year abatement plan updated every year). The contractor shall conduct ACM condition inspections as required. The contractor shall plan and program asbestos abatement projects to correct deficiencies found. Asbestos abatement should be incorporated into facility maintenance, repair or renovation projects where feasible. The contractor shall submit a 5-year abatement plan updated every year to NASA-KSC and the 45th Space Wing asbestos program managers and infrastructure maintenance managers. The contractor shall remove ACM for maintenance, operations, and mission requirements in accordance with all Federal, State, local, Air Force regulations and requirements, and as described in Appendix "A" of Section "J", Attachment "J-1". **(Mod 397)**

When health hazards are identified, ACM is being disturbed because of modification activities, or ACM is identified as deteriorated by any source, the contractor shall abate the asbestos, re-insulating, or resurfacing as needed.

2.2.1.15 Utility Coordination, Reports and Generator Inventories. The contractor shall support activities required to manage utilities. Utility coordination activities shall include the notification of impacts and scheduling utility outages for NASA and CCAFS. Utility coordination activities shall also include the provision of the CCAFS Unscheduled Utility Outage and Incident Reports specified in accordance with DRD 1.1-04, Report Anomaly and the NASA Quick-Look Unscheduled Utility Outage Incident Reports, on an as-needed basis. Using the USAF Real Property records and the O/M/E/U Matrix, the contractor shall provide an annual listing of CCAFS/PAFB generators in support of the USAF Annual Generator Inventories.

The contractor shall communicate/coordinate maintenance and operations with FPL thru the NASA Operating Representative which is contained in the compliance document "Interconnection Agreement Between Florida Power & Light Company and the National Aeronautics and Space Administration." **(Mod 299)**

2.2.1.16 Dredge and Spoil Areas. The contractor shall mow the embankments at least once every 6 months of 3 government dredge and spoil areas. This support is limited to mowing the slopes as well as the center track. The extent of this mowing is as indicated below:

- Dike 2B is 7,100 linear feet long
- Dike 3 is 5,500 linear feet long
- Dike 5 is 4,400 linear feet long

2.2.1.17 Support to Vessels. The contractor shall provide support to home-ported and transient vessels at the Trident Poseidon Wharfs and the Army Outport on CCAFS. This support includes electrical and water waste connections and dockside services for arriving/departing vessels. The contractor shall also provide minor ship repairs, as directed. These actions include minor electrical, metal or carpentry work not associated with vessel structural integrity or seaworthiness.

The contractor shall provide management, certification, operation, and maintenance of dockside portal and fixed cranes inside CCAFS Naval facilities, in accordance with NAVFAC P-307. Additionally, hazardous waste shall be received from ships and processed for disposal.

The contractor shall provide an OSHA-compliant oil spill recovery team and maintain all oil spill recovery equipment, to include booms, buoys, anchors, deployment boats, and absorption materials, to ensure availability on immediate notice.

Facility Condition Assessments of NOTU FSEU shall be conducted in accordance with NAVFAC MO-322, Volume 1. Supporting documentation for the preparation of the annual NOTU FSEU O&M budget shall be prepared in accordance with OPNAVINST 11010.34.

2.2.1.18 Deleted.

2.2.1.19 Clean Rooms. The contractor shall ensure that NASA and Air Force clean rooms and clean work area facilities and associated support equipment at CCAFS meet customer cleanliness requirements (Facility numbers 01428, 01619, 01728, 49635, 55005, 60505, and 60680). The contractor shall use engineering and personnel controls, housekeeping procedures, and contamination-generating constraints to monitor and maintain high levels of cleanliness in the Contractor Work Authorizations (CWAs) assigned to J-BOSC. The contractor shall implement all procedures specified in referenced compliance documents. Description and classification of clean rooms are contained in Technical Exhibits 5.5-801 and 5.5-802.

The contractor shall use engineering and personnel controls, housekeeping procedures, and contamination-generating constraints to monitor and maintain high levels of cleanliness in clean work areas. The contractor shall implement all procedures specified in KCI-HB-5340.1, Payload Facility Contamination Control Plan. In addition, the contractor shall apply the standard and alternate classes of air cleanliness for clean rooms and clean zones based on specified concentrations of airborne particles, in accordance with FED-STD 209E, Federal Standard Airborne Particulate Cleanliness Classes in Clean Rooms and Clean Zones.

2.2.1.20 Deleted.

2.2.1.21 Special Dewatering Support. The contractor shall provide support for storage, treatment, or disposal of dewatering effluent as a result of construction, manhole entry, or other events requiring dewatering. (CCR 06-44, **Mod 398**)

2.2.2 Refuse, Pest Control and Roads/Grounds Maintenance

The contractor shall provide grounds and landscape maintenance for KSC as listed in Technical Exhibit. 7.0-016. The contractor shall provide grounds and landscape maintenance for CCAFS, and the Florida Annexes as listed in Tech. Exh. 5.5-826. Contractor responsibilities shall include: grass mowing, edging or trimming; eliminating weeds; applying fertilizer; maintaining the landscaping; cleaning up debris; watering; and maintaining the watering system. The contractor shall provide sanitation (trash pick up and disposal) in accordance with Technical Exhibits 2.2.2.3A Florida Annexes Refuse Collection Sites and Frequencies, 2.2.2.3B CCAFS Refuse Collection Sites and Frequencies, and 2.2.2.3C KSC Refuse Collection Sites and Frequencies. (The contractor shall consolidate refuse collection and disposal contracts to realize savings from economy of scale. To comply with state permit requirements, one landfill operator shall be on duty at all times that the landfill is open for disposal activities. This operator shall physically witness all tippage and shall sign off on manifests as appropriate. **(Mod 269)**) The contractor shall provide pest control services for assigned facilities, systems, and equipment as listed in the OMEU Matrix. Roads Maintenance includes inspection, maintenance, and repair of ditches and storm water systems and the operation of bridges on KSC. **(Mod 469)**

2.2.2.1 Roads and Grounds. The contractor shall:

- Perform operations, maintenance, and engineering (O/M/E) on all roads, airfields, parking lots, pavement, and bridges on KSC and CCAFS, including marking and street sweeping;
- Provide engineering to analyze and make recommendations for road rehabilitation contracts. Contractor shall sweep, vacuum, and maintain the concrete SLF and Skid Strip runway and parking aprons, and the SLF tow-way to K6-894 (OM) and K6-848 (VAB) to maintain an environment that is free of foreign object debris (FOD) and ready for Orbiter landing and towing operations;
- Provide annual inspections of all paved roadways to identify areas that require repair following the guidance of the Florida Department of Transportation Manual of Uniform Minimum Standards for Design, Construction and Maintenance for Streets and Highways (Florida Green Book, May 2002); and
- Provide the annual inspection results, within 30 working days following the annual inspection, to the CCSMO.
- The contractor shall provide roads maintenance at KSC, CCAFS, and the Florida Annexes following the guidance in paragraph 10 of the Florida Department of Transportation Green Book. This shall include inspection and maintenance of pavement and shoulders, with particular emphasis on maintaining shoulders flush with the pavement, and debris removal

from the pavement, shoulders, and roadside clear zones. The contractor shall provide operator coverage 24 hrs per day, 7 days a week for the NASA Causeway (Indian River) and Haulover Canal drawbridges and shall operate the NASA Causeway (Banana River) drawbridge and Jay-Jay Railroad Bridge on an on-call basis. Roads maintenance areas of responsibility for KSC and CCAFS are listed in Technical Exhibit 2.2.2.1-01. The contractor shall be responsible for all areas of Florida Annexes.

The contractor shall operate and maintain Government Furnished Equipment (GFE) provided for Roads and Grounds support and listed under the NASA Equipment Management System database. Operators shall be properly licensed and equipment shall be maintained in safe and reliable condition.

The contractor shall maintain a designated Memorial Tree Planting Area east of the Headquarters Building. This task includes pruning trees, shrubs, and hedges and removing the cuttings, replacing grass sod, and removing stumps and dead plants. The contractor shall maintain grass, shrubs, and trees in a healthy condition.

The contractor shall maintain existing "cleared" Lines of Sight (LOS), fence lines and utility corridors in conformance with Technical Exhibit 5.5-827. Contractor is expected to maintain approximately 250 acres of LOS and 450 acres of utility corridors under this contract. LOS includes Range, Public Affairs (VIP/press site views) and navigational Aid Range Markers. The contractor shall also provide initial clearing of LOS, fence lines and utility corridors as Type 3C projects. **(Mod 293) (Mod 469)**

2.2.2.2 Pest Control. The contractor shall provide pest control services associated to structures, on a trouble call basis, for KSC, CCAFS, and the Florida Annexes facilities listed in the O/M/E/U Matrix where contractor maintenance responsibilities include structures. The contractor shall develop and implement a pest control program for Cape Canaveral Spaceport. All buildings, facilities, and outside work areas shall be inspected and sprayed at a frequency necessary to prevent damage to structures and control pests that may affect health and morale. The contractor shall provide monthly inspections and treatment, as needed, to all food preparation and service areas of KSC and CCAFS. Food preparation and service areas include the ITL Warehouse cafeteria, E&L Shoppette, Cape Main Cafeteria, and Green House Navy Cafeteria at CCAFS. For KSC food preparation and service areas, included are areas located at O&C, SSPF, HQ, CIF, ARF, MFF, OSB, LCC, Conference Center, Lackmann Warehouse, and USA Logistics. **(Mod 451)**

The contractor shall trap and relocate small nuisance mammals that are in or on a structure on KSC and CCAFS. Mammals that have had direct human contact (bite or scratch) shall be turned over to animal control or the health department for rabies determination. **(Mod 451)**

The contractor shall comply with The Air Force Pest Management Plan in accordance with Technical Exhibit 2.2.2.2-01. The following items, pertaining to Technical Exhibit 2.2.2.2-01 entries, Mosquito Fogging/Larvae Control shall be provided to all areas of CCAFS. Termite Treatment/Inspection shall not be provided for Cape Canaveral Air Force Station. For all areas of Kennedy Space Center and NASA facilities on CCAFS, mosquito fogging/larvae control and termite treatment/inspection services will continue to be provided. **(Mod 451)**

2.2.2.3 Trash Collection and Disposal. The contractor shall collect and dispose of refuse and bulk items for KSC and CCAFS. For the Florida Annexes, the contractor shall only collect, transport, and dispose of refuse. The contractor shall provide special refuse handling at CCAFS and KSC for International Regulated Food Waste and trash resulting from food products obtained outside the United States, and shall operate the CCAFS Construction and Demolition Debris Landfill (C&D), CCAFS Asbestos Monofill, and KSC Class III landfills. This refuse task includes providing services to empty compactors and containers for CCAFS and KSC and